

Juan Pablo Carvajal González

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Research Profile

Observational astrophysicist studying the evolution of dwarf galaxies in groups and clusters. I combine VLT/MUSE integral-field spectroscopy with H I, molecular-gas, ultraviolet, infrared, and optical data to reconstruct gas removal, star-formation suppression, kinematics, and chemical evolution. My broader work also spans spectroscopy and photometry of solar system minor planets and interstellar visitors.

Education

Ph.D. in Astrophysics

Mar 2023–Feb 2027
(expected)

Pontificia Universidad Católica de Chile, Santiago

GPA: **6.9/7.0**. Thesis: *Environmental Processing of First-Infall Dwarf Galaxies in the Fornax Cluster: A MUSE Perspective*.

Adviser: Prof. Thomas H. Puzia.

B.Sc. in Astronomy

Feb 2017–Nov 2022

Universidad de Antioquia, Medellín, Colombia

GPA: **4.44/5.00**. Undergraduate thesis: *Towards the Systematic Detection of Active Asteroid Candidates: A Photometric Method*. Adviser: Prof. Juan Carlos Muñoz-Cuertas.

Principal-Investigator Observing Programmes

VLT/MUSE — Stellar Stoppage: The Fornax Cluster's Influence on First-Time Visitor Dwarf Galaxies 2025–2026
116.290B.001

PI; spatially resolved spectroscopy of 13 (seven executed) cold-gas-rich, environmentally disturbed Fornax dwarfs. 13.6 h of on-target integration.

VLT/FORS2 — Direct Abundances as a Test of Environment-Driven Chemical Offsets in Fornax Dwarfs 2026
117.2AY1.001

PI; **21 h awarded** for blue MXU spectroscopy of H II regions in five dwarfs.

APEX/nFLASH230 — Searching for Warm, Displaced Molecular Gas in NGC 1427A with APEX CO(2–1) 2026
C-0117.F-9706C-2026

PI; **25 h awarded** through CNTAC for a five-pointing molecular-gas search; observations completed.

Refereed Publications and Accepted Articles

1. **J. P. Carvajal**, Puzia, T. H., Blaña, M., et al. (2026). *Tidal pre-conditioning and ram-pressure stripping in NGC 1427A: Deep VLT/MUSE spectroscopy and FUV-to-radio observations trace a Fornax Cluster dwarf in transformation*. *A&A*, **712**, A29. doi:10.1051/0004-6361/202659045.
2. Fahrion, K., Hilker, M., Chaturvedi, A., **J. P. Carvajal**, & Puzia, T. H. (2026). *The complex history of NGC 1427A revealed by its star clusters and star formation history*. *A&A*, **712**, A28. doi:10.1051/0004-6361/202659061.
3. Luco, B., Puzia, T. H., Rahatgaonkar, R., **J. P. Carvajal**, & Nayak, P. K. (2026). *Very Large Telescope observations of interstellar comet 3I/ATLAS III: High-resolution monitoring of CN and forbidden oxygen emission across the perihelion passage with ESPRESSO*. *Accepted in A&A*. arXiv:2607.15355.
4. Chandler, C. O., Bernardinelli, P. H., Jurić, M., et al. (Rubin Observatory team; including **J. P. Carvajal**) (2026). *NSF–DOE Vera C. Rubin Observatory observations of interstellar comet 3I/ATLAS (C/2025 N1)*. *ApJL*, **1001**, L35. doi:10.3847/2041-8213/ae4b3a.
5. Hutsemékers, D., Manfroid, J., et al. (including **J. P. Carvajal**) (2026). *Pre-perihelion evolution of the Ni I/Fe I abundance ratio in the coma of the interstellar comet 3I/ATLAS: From extreme to normal*. *A&A*, **706**, A43. doi:10.1051/0004-6361/202557484.
6. Ordenes-Briceño, Y., Puzia, T. H., Eigenthaler, P., Blaña, M., **J. P. Carvajal**, et al. (2026). *The Next Generation Fornax Survey (NGFS). VIII. A support vector machine approach to disentangling globular clusters*. *A&A*, **706**, A368. doi:10.1051/0004-6361/202556455.

7. Rahatgaonkar, R., **J. P. Carvajal**, Puzia, T. H., et al. (2025). *Very Large Telescope observations of interstellar comet 3I/ATLAS. II. From quiescence to glow: Dramatic rise of Ni I emission and incipient CN outgassing at large heliocentric distances*. *ApJL*, **995**, L34. doi:10.3847/2041-8213/ae1cbc.
8. Puzia, T. H., Rahatgaonkar, R., **J. P. Carvajal**, Nayak, P. K., & Luco, B. (2025). *Spectral characteristics of interstellar object 3I/ATLAS from SOAR observations*. *ApJL*, **990**, L27. doi:10.3847/2041-8213/adfa0b.
9. **J. P. Carvajal**, Bauer, F. E., Reyes-Jainaga, I., et al. (2025). *Tuning into the spatial frequency space: Satellite and space debris detection in the ZTF alert stream*. *A&A*, **697**, A119. doi:10.1051/0004-6361/202452880.
10. **J. P. Carvajal** & Muñoz-Cuartas, J. C. (2023). *Towards the systematic detection of active asteroid candidates: A photometric method*. *MNRAS*, **521**, 6075–6088. doi:10.1093/mnras/stad897.

Work in Preparation and Other Research Outputs

- **J. P. Carvajal** et al. *Stellar stoppage in the Fornax cluster: MUSE constraints on the gas-to-quiescence transition of dwarf galaxies*. First-author manuscript in preparation; homogeneous analysis of a 38-dwarf MUSE sample.
- **J. P. Carvajal**, Puzia, T. H., Galaz, G., Ordenes-Briceño, Y., Eigenthaler, P., & Johnston, E. (2026). *Unveiling intermediate-redshift galaxy clusters within the Next Generation Fornax Survey*. *RMxAC*, **60**, 258–278; XVII Latin American Regional IAU Meeting proceedings.
- **J. P. Carvajal** et al. (2026). *Deep MUSE Mosaic of NGC 1427A in the Fornax Cluster: reduced and calibrated mosaic cube and white-light image*. ESO Science Archive Phase 3 collection, doi:10.18727/archive/109.

Honors and Fellowships

First Place, Three Minute Thesis (3MT[®]) UC 2026 , Pontificia Universidad Católica de Chile; selected to represent UC in the Universitas 21 international competition.	2026
Second Place, Natural Sciences , 27th National Otto de Greiff Competition for Best Undergraduate Theses, Colombia.	2025
National Doctoral Scholarship , Agencia Nacional de Investigación y Desarrollo (ANID), Chile.	2024
Best Graduate of the 2022 Astronomy cohort , Universidad de Antioquia; recipient of the university postgraduate-study scholarship.	2023
Best Academic Student in Astronomy (three distinctions), Universidad de Antioquia.	2017, 2018, 2022
Bronze Medal , Third Latin American and Caribbean University Physics Olympiad, Havana, Cuba.	2019

Selected Research Talks

<i>Early Environmental Transformation of Dwarf Galaxies in the Fornax Cluster</i> , research seminar, Universidad Andrés Bello, Santiago.	Aug 2026
<i>Early Environmental Transformation of Dwarf Galaxies in the Fornax Cluster: Insights from NGC 1427A</i> , Thirty Minute Talk, European Southern Observatory, Santiago.	May 2026
<i>Early Environmental Transformation of Dwarf Galaxies in the Fornax Cluster: Insights from NGC 1427A</i> , COSMOS Seminar, Department of Astronomy, University of Texas at Austin.	Apr 2026

Selected Conference Presentations

<i>Probing Galaxy Evolution with the Next Generation Fornax Survey at $z < 1$</i> , poster, LSST@LATAM, Universidad de La Serena.	Jun 2024
<i>Unveiling Intermediate-Redshift Galaxy Clusters in the Fornax Field with NGFS</i> , poster, XVII Latin American Regional IAU Meeting (LARIM), Montevideo, Uruguay.	Nov 2023

Teaching and Mentoring

Teaching Assistant, AST0112 Astronomy

Aug–Dec 2025

Pontificia Universidad Católica de Chile

Supported an English-language introductory astronomy course through problem-solving tutorials, review materials, assessment preparation, grading support, and student feedback.

Peer Academic Tutor

Oct 2020–Oct 2021

Universidad de Antioquia

Volunteered through the Pares Académicos Solidarios programme, tutoring individuals and small groups in introductory physics and supporting first-year students' transition to university study.

Pedagogical training: Teaching Skills Development Workshop (VRI0007), UC Chile, 2024; Teaching Assistant Training Programme, UC Chile, 2025.

Academic Service and Outreach

Organizing Committee Member, RECA Education / BARCo

Jan 2024–Jun 2025

Red de Estudiantes Colombianos en Astronomía

Co-designed hands-on astronomy activities and offline-friendly teaching materials for rural Colombian schools; coordinated with participating schools and supported teachers in implementing the programme. BARCo was funded by the IAU Office of Astronomy for Development.

Administrative Assistant

Mar 2024–Jan 2026

Graduate School, Pontificia Universidad Católica de Chile

Supported graduate-school events and student communications and prepared statistical summaries and feedback reports for academic programmes.

Research Methods, Software, and Languages

Observational methods: optical integral-field spectroscopy; stellar-population and full-spectrum fitting; stellar and ionized-gas kinematics; emission-line and abundance diagnostics; reduction and analysis of FUV-to-radio observations; observing-proposal design and Phase 2 preparation.

Software and workflows: Python, Astropy, MPDAF, pPXF, scikit-learn, C language, ESO pipelines/EsoRex, Git, Bash, L^AT_EX, etc.

Languages: Spanish (native); English (proficient; TOEFL iBT 104/120).